COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31S2
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ACTIVITY NO. 6.1 **Targeting Specific Nodes**

1. PROCEDURE OUTPUT

Task 1: Targeting Specific Nodes

1. Create a new playbook and named it site.yml. Follow the commands as shown in the image below. Make sure to save the file and exit.

```
  GNU nano 6.2                                site.yml
---

- hosts: all
  become: true
  tasks:

  - name: install apache and php for Ubuntu servers
    apt:
      name:
        - apache2
        - libapache2-mod-php
      state: latest
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

    - name: install apache and php for CentOS servers
      dnf:
        name:
          - httpd
          - php
        state: latest
      when: ansible_distribution == "CentOS"
```

2. Edit the inventory file. Remove the variables we put in our last activity and group according to the image shown below:

```
  GNU nano 6.2
[web_servers]
192.168.56.9 ansible_user=queleroyubuntu3

[db_servers]
192.168.56.10 ansible_user=queleroyubuntu3

[file_servers]
192.168.56.12 ansible_user=leroyq
```

Make sure to save the file and exit.
Right now, we have created groups in our inventory file and put each server in its own group. In other cases, you can have a server be a member of multiple groups, for example you have a test server that is also a web server.

3. Edit the site.yml by following the image below:

```
  GNU nano 6.2                          site.yml *
---

- hosts: all
  become: true
  pre_tasks:

  - name: install updates (CentOS)
    dnf:
      update_only: yes
      update_cache: yes
    when: ansible_distribution == "CentOS"

  - name: install updates (Ubuntu)
    apt:
      upgrade: dist
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

- hosts: web_servers
  become: true
  tasks:

  - name: install apache and php for Ubuntu servers
    apt:
      name:
        - apache2
        - libapache2-mod-php
      state: latest
      update_cache: yes
```

Make sure to save the file and exit.

The pre-tasks command tells the ansible to run it before any other thing. In the

pre-tasks, CentOS will install updates while Ubuntu will upgrade its distribution package. This will run before running the second play, which is targeted at web_servers. In the second play, apache and php will be installed on both Ubuntu servers and CentOS servers.

```
queleroyubuntu3@workstation:~/CPE212_QUE$ ansible-playbook -i inventory.ini site.yml
BECOME password:

PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.9]
ok: [192.168.56.10]
[DEPRECATION WARNING]: Distribution centos 9 on host 192.168.56.12 should use
/usr/libexec/platform-python, but is using /usr/bin/python for backward
compatibility with prior Ansible releases. A future Ansible release will
default to using the discovered platform python for this host. See https://docs
.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for
more information. This feature will be removed in version 2.12. Deprecation
warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
ok: [192.168.56.12]

TASK [install updates (CentOS)] ************************************************
skipping: [192.168.56.9]
skipping: [192.168.56.10]
ok: [192.168.56.12]

TASK [install updates (Ubuntu)] ************************************************
skipping: [192.168.56.12]
fatal: [192.168.56.9]: FAILED! => {"changed": false, "msg": "'/usr/bin/apt-get dist-upgrade ' failed: E: Could not get lock /var/lib
/dpkg/lock-frontend. It is held by process 3088 (unattended-upgr)\nE: Unable to acquire the dpkg frontend lock (/var/lib/dpkg/lock-f
rontend), is another process using it?\n", "rc": 100, "stdout": "", "stdout_lines": []}

changed: [192.168.56.10]

PLAY [web_servers] *************************************************************

PLAY RECAP *********************************************************************
192.168.56.10              : ok=2    changed=1    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.12              : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.9               : ok=1    changed=0    unreachable=0    failed=1    skipped=1    rescued=0    ignored=0
```

Run the site.yml file and describe the result.

- Ansible is able to perform the job of installing the updates in the computers that have the necessary requirements only. The playbook provides for that task of installing the Apache and PHP only on the computers belonging to the web_server group.

4. Let's try to edit again the site.yml file. This time, we are going to add plays targeting the other servers. This time we target the db_servers by adding it on the current site.yml. Below is an example: (Note add this at the end of the playbooks from task 1.3.

Make sure to save the file and exit.

```
ok: [192.168.56.12]

TASK [install updates (CentOS)] ************************************************************************************************************
skipping: [192.168.56.9]
skipping: [192.168.56.10]
ok: [192.168.56.12]

TASK [install updates (Ubuntu)] ************************************************************************************************************
skipping: [192.168.56.12]
ok: [192.168.56.9]
ok: [192.168.56.10]

PLAY [web_servers] *************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************
ok: [192.168.56.9]

TASK [install apache and php for Ubuntu servers] *******************************************************************************************
ok: [192.168.56.9]

TASK [install apache and php for CentOS servers] *******************************************************************************************
skipping: [192.168.56.9]

PLAY [db_servers] **************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************
ok: [192.168.56.10]

TASK [install mariadb package (CentOS)] ****************************************************************************************************
skipping: [192.168.56.10]

TASK [Mariadb- Restarting/Enabling] ********************************************************************************************************
fatal: [192.168.56.10]: FAILED! => {"changed": false, "msg": "Could not find the requested service mariadb: host"}

PLAY RECAP *********************************************************************************************************************************
192.168.56.10              : ok=3    changed=0    unreachable=0    failed=1    skipped=2    rescued=0    ignored=0
192.168.56.12              : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.9               : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

Run the site.yml file and describe the result.

- The playbook has an additional task that is executed for the computers belonging to the db_server group. The MariaDB package has been successfully installed regardless of the fact that the OS distribution of some computers did not meet the requirements.

5. Go to the remote server (Ubuntu) terminal that belongs to the db_servers group and check the status for mariadb installation using the command: systemctl status mariadb. Do this on the CentOS server also.

```
queleroyubuntu3@server1:~$ systemctl status mariadb
● mariadb.service - MariaDB 10.6.23 database server
     Loaded: loaded (/lib/systemd/system/mariadb.service; enabled; vendor preset: enabled)
     Active: active (running) since Thu 2026-09-03 09:48:24 CST; 26s ago
       Docs: man:mariadbd(8)
             https://mariadb.com/kb/en/library/systemd/
    Process: 36857 ExecStartPre=/usr/bin/install -m 755 -o mysql -g root -d /var/run/mysqld (code=exited, status=0/SUCCESS)
    Process: 36859 ExecStartPre=/bin/sh -c systemctl unset-environment _WSREP_START_POSITION (code=exited, status=0/SUCCESS)
    Process: 36861 ExecStartPre=/bin/sh -c [ ! -e /usr/bin/galera_recovery ] && VAR= ||   VAR=`/usr/bin/galera_recovery`; [ $? -eq
    Process: 36901 ExecStartPost=/bin/sh -c systemctl unset-environment _WSREP_START_POSITION (code=exited, status=0/SUCCESS)
    Process: 36903 ExecStartPost=/etc/mysql/debian-start (code=exited, status=0/SUCCESS)
   Main PID: 36890 (mariadbd)
     Status: "Taking your SQL requests now..."
      Tasks: 12 (limit: 30342)
     Memory: 61.3M
        CPU: 767ms
     CGroup: /system.slice/mariadb.service
             └─36890 /usr/sbin/mariadbd
lines 1-17/17 (END)
```

```
queleroyubuntu3@workstation:~/CPE212_QUE$ ssh leroyq@192.168.56.12
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Sep  3 09:48:23 2026 from 192.168.56.11
[leroyq@localhost ~]$ systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
     Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
     Active: active (running) since Thu 2026-09-03 09:48:24 PST; 1min 27s ago
       Docs: man:mariadbd(8)
             https://mariadb.com/kb/en/library/systemd/
    Process: 17810 ExecStartPre=/usr/libexec/mariadb-check-socket (code=exited, status=0/SUCCESS)
    Process: 17833 ExecStartPre=/usr/libexec/mariadb-prepare-db-dir mariadb.service (code=exited, status=0/SUCCESS)
    Process: 17888 ExecStartPost=/usr/libexec/mariadb-check-upgrade (code=exited, status=0/SUCCESS)
   Main PID: 17868 (mariadbd)
     Status: "Taking your SQL requests now..."
      Tasks: 9 (limit: 22978)
     Memory: 65.9M (peak: 68.1M)
        CPU: 360ms
     CGroup: /system.slice/mariadb.service
             └─17868 /usr/libexec/mariadbd --basedir=/usr

Sep 03 09:48:24 localhost.localdomain systemd[1]: Starting MariaDB 10.5 database server...
Sep 03 09:48:24 localhost.localdomain mariadb-prepare-db-dir[17833]: Database MariaDB is probably initialized in /var/lib/mysql alr>
Sep 03 09:48:24 localhost.localdomain mariadb-prepare-db-dir[17833]: If this is not the case, make sure the /var/lib/mysql is empty>
Sep 03 09:48:24 localhost.localdomain systemd[1]: Started MariaDB 10.5 database server.
lines 1-20/20 (END)
```

Describe the output.

- The status output from systemctl confirms that the MariaDB service is still running on both Ubuntu and CentOS computer nodes.

6. Edit the site.yml again. This time we will append the code to configure installation on the file_servers group. We can add the following on our file.

Make sure to save the file and exit.

```
PLAY [web_servers] ************************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************************
ok: [192.168.56.10]

TASK [install apache and php for Ubuntu servers] ******************************************************************************************
ok: [192.168.56.10]

TASK [install apache and php for CentOS servers] ******************************************************************************************
skipping: [192.168.56.10]

PLAY [db_servers] *************************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************************
ok: [192.168.56.9]

TASK [install mariadb package (CentOS)] ***************************************************************************************************
skipping: [192.168.56.9]

TASK [install mariadb package (Ubuntu)] ***************************************************************************************************
ok: [192.168.56.9]

TASK [Mariadb- Restarting/Enabling] *******************************************************************************************************
changed: [192.168.56.9]

PLAY [file_servers] ***********************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************************
ok: [192.168.56.12]

TASK [install samba package] **************************************************************************************************************
changed: [192.168.56.12]

PLAY RECAP ********************************************************************************************************************************
192.168.56.10              : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.12              : ok=4    changed=1    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.9               : ok=5    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

Run the site.yml file and describe the result.

- Ansible has run the playbook three times, the first one is executed on web_servers and db_servers machines while the third one is executed on the computers belonging to the file_server group.

The testing of the file_servers is beyond the scope of this activity, and as well as our topics and objectives. However, in this activity we were able to show that we can target hosts or servers using grouping in ansible playbooks.

Task 2: Using Tags in running playbooks

In this task, our goal is to add metadata to our plays so that we can only run the plays that we want to run, and not all the plays in our playbook.

1. Edit the site.yml file. Add tags to the playbook. After the name, we can place the tags: name_of_tag. This is an arbitrary command, which means you can use any name for a tag.

Make sure to save the file and exit.

```
PLAY [web_servers] *************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************
ok: [192.168.56.10]

TASK [install apache and php for Ubuntu servers] *******************************************************************************
ok: [192.168.56.10]

TASK [install apache and php for CentOS servers] *******************************************************************************
skipping: [192.168.56.10]

PLAY [db_servers] **************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************
ok: [192.168.56.9]

TASK [install mariadb package (CentOS)] ****************************************************************************************
skipping: [192.168.56.9]

TASK [install mariadb package (Ubuntu)] ****************************************************************************************
ok: [192.168.56.9]

TASK [Mariadb- Restarting/Enabling] ********************************************************************************************
changed: [192.168.56.9]

PLAY [file_servers] ************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************
ok: [192.168.56.12]

TASK [install samba package] ***************************************************************************************************
ok: [192.168.56.12]

PLAY RECAP *********************************************************************************************************************
192.168.56.10              : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.12              : ok=4    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.9               : ok=5    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

Run the site.yml file and describe the result.

2. On the local machine, try to issue the following commands and describe each result:

2.1 ansible-playbook --list-tags site.yml

```
queleroyubuntu3@workstation:~/CPE212_QUE$ ansible-playbook --list-tags site.yml

playbook: site.yml

  play #1 (all): all    TAGS: []
      TASK TAGS: [always]

  play #2 (web_servers): web_servers    TAGS: []
      TASK TAGS: [apache, apache2, centos, httpd, ubuntu]

  play #3 (db_servers): db_servers      TAGS: []
      TASK TAGS: [centos, db, mariadb, ubuntu]

  play #4 (file_servers): file_servers  TAGS: []
      TASK TAGS: [samba]
```

- This command shows the general overview of all plays along with the task tags such as apache, db and samba without executing any tasks on the hosts.

2.2 ansible-playbook --tags centos --ask-become-pass site.yml

```
deprecation_warnings=False in ansible.cfg.
ok: [192.168.56.12]

TASK [install updates (CentOS)] ************************************************************************************************************
skipping: [192.168.56.10]
skipping: [192.168.56.9]
ok: [192.168.56.12]

TASK [install updates (Ubuntu)] ************************************************************************************************************
skipping: [192.168.56.12]
ok: [192.168.56.9]
ok: [192.168.56.10]

PLAY [web_servers] *************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************
ok: [192.168.56.10]

TASK [install apache and php for CentOS servers] *******************************************************************************************
skipping: [192.168.56.10]

PLAY [db_servers] **************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************
ok: [192.168.56.9]

TASK [install mariadb package (CentOS)] ****************************************************************************************************
skipping: [192.168.56.9]

PLAY [file_servers] ************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************
ok: [192.168.56.12]

PLAY RECAP *********************************************************************************************************************************
192.168.56.10              : ok=3    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.12              : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.9               : ok=3    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

- Ansible will execute only those tasks that carry the centos tag for every play and ignore any configuration tasks related to Ubuntu.

2.3 ansible-playbook --tags db --ask-become-pass site.yml

```
deprecation_warnings=False in ansible.cfg.
ok: [192.168.56.12]

TASK [install updates (CentOS)] ************************************************************************************************************
skipping: [192.168.56.10]
skipping: [192.168.56.9]
ok: [192.168.56.12]

TASK [install updates (Ubuntu)] ************************************************************************************************************
skipping: [192.168.56.12]
ok: [192.168.56.9]
ok: [192.168.56.10]

PLAY [web_servers] *************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************
ok: [192.168.56.10]

PLAY [db_servers] **************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************
ok: [192.168.56.9]

TASK [install mariadb package (CentOS)] ****************************************************************************************************
skipping: [192.168.56.9]

TASK [install mariadb package (Ubuntu)] ****************************************************************************************************
ok: [192.168.56.9]

PLAY [file_servers] ************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************
ok: [192.168.56.12]

PLAY RECAP *********************************************************************************************************************************
192.168.56.10              : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.12              : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.9               : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

- Ansible executes only those tasks that have a db tag meaning that database packages will be installed while web and file server tasks will be skipped.

2.4 ansible-playbook --tags apache --ask-become-pass site.yml

```
deprecation_warnings=False in ansible.cfg.
ok: [192.168.56.12]

TASK [install updates (CentOS)] ************************************************************************************************************
skipping: [192.168.56.10]
skipping: [192.168.56.9]
ok: [192.168.56.12]

TASK [install updates (Ubuntu)] ************************************************************************************************************
skipping: [192.168.56.12]
ok: [192.168.56.9]
ok: [192.168.56.10]

PLAY [web_servers] *************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************
ok: [192.168.56.10]

TASK [install apache and php for Ubuntu servers] *******************************************************************************************
ok: [192.168.56.10]

TASK [install apache and php for CentOS servers] *******************************************************************************************
skipping: [192.168.56.10]

PLAY [db_servers] **************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************
ok: [192.168.56.9]

PLAY [file_servers] ************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************
ok: [192.168.56.12]

PLAY RECAP *********************************************************************************************************************************
192.168.56.10              : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.12              : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.9               : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

- In the web_servers play, the only executed tasks are those with an apache tag while database and file server configurations won't be executed.

2.5 ansible-playbook --tags “apache,db” --ask-become-pass site.yml

```
ok: [192.168.56.12]

TASK [install updates (Ubuntu)] ************************************************************************************************
skipping: [192.168.56.12]
ok: [192.168.56.9]
ok: [192.168.56.10]

PLAY [web_servers] *************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************
ok: [192.168.56.10]

TASK [install apache and php for Ubuntu servers] *******************************************************************************
ok: [192.168.56.10]

TASK [install apache and php for CentOS servers] *******************************************************************************
skipping: [192.168.56.10]

PLAY [db_servers] **************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************
ok: [192.168.56.9]

TASK [install mariadb package (CentOS)] ****************************************************************************************
skipping: [192.168.56.9]

TASK [install mariadb package (Ubuntu)] ****************************************************************************************
ok: [192.168.56.9]

PLAY [file_servers] ************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************
ok: [192.168.56.12]

PLAY RECAP *********************************************************************************************************************
192.168.56.10              : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.12              : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.9               : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

- Ansible produces an output with tasks with tags either apache or db executed smoothly while other tasks not tagged such as installation of samba will be skipped.

Task 3: Managing Services

1. Edit the file site.yml and add a play that will automatically start the httpd on CentOS server.
Figure 3.1.1
Make sure to save the file and exit.
You would also notice from our previous activity that we already created a module that runs a service.
Figure 3.1.2
This is because in CentOS, installed packages’ services are not run automatically. Thus, we need to create the module to run it automatically.

2. To test it, before you run the saved playbook, go to the CentOS server and stop the currently running httpd using the command sudo systemctl stop httpd. When prompted, enter the sudo password. After that, open the browser and enter the CentOS server’s IP address. You should not be getting a display because we stopped the httpd service already.

3. Go to the local machine and this time, run the site.yml file. Then after running the file, go again to the CentOS server and enter its IP address on the browser.

```
TASK [install apache and php for CentOS servers] ******************************
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to `True` is deprecated, top-level facts will not be auto injected after the change. This feature will be removed from ansible-core version 2.24.
Origin: /home/leroy1/CPE212_QUE/site.yml:44:11

42         - php
43       state: latest
44     when: ansible_distribution == "CentOS"
             ^ column 11

Use `ansible_facts["fact_name"]` (no `ansible_` prefix) instead.

skipping: [192.168.56.104]
ok: [192.168.56.107]

TASK [start httpd (CentOS)] ***************************************************
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to `True` is deprecated, top-level facts will not be auto injected after the change. This feature will be removed from ansible-core version 2.24.
Origin: /home/leroy1/CPE212_QUE/site.yml:52:11

50       state: started
51       enabled: true
52     when: ansible_distribution == "CentOS"
             ^ column 11

Use `ansible_facts["fact_name"]` (no `ansible_` prefix) instead.

skipping: [192.168.56.104]
changed: [192.168.56.107]

PLAY [db_servers] *************************************************************

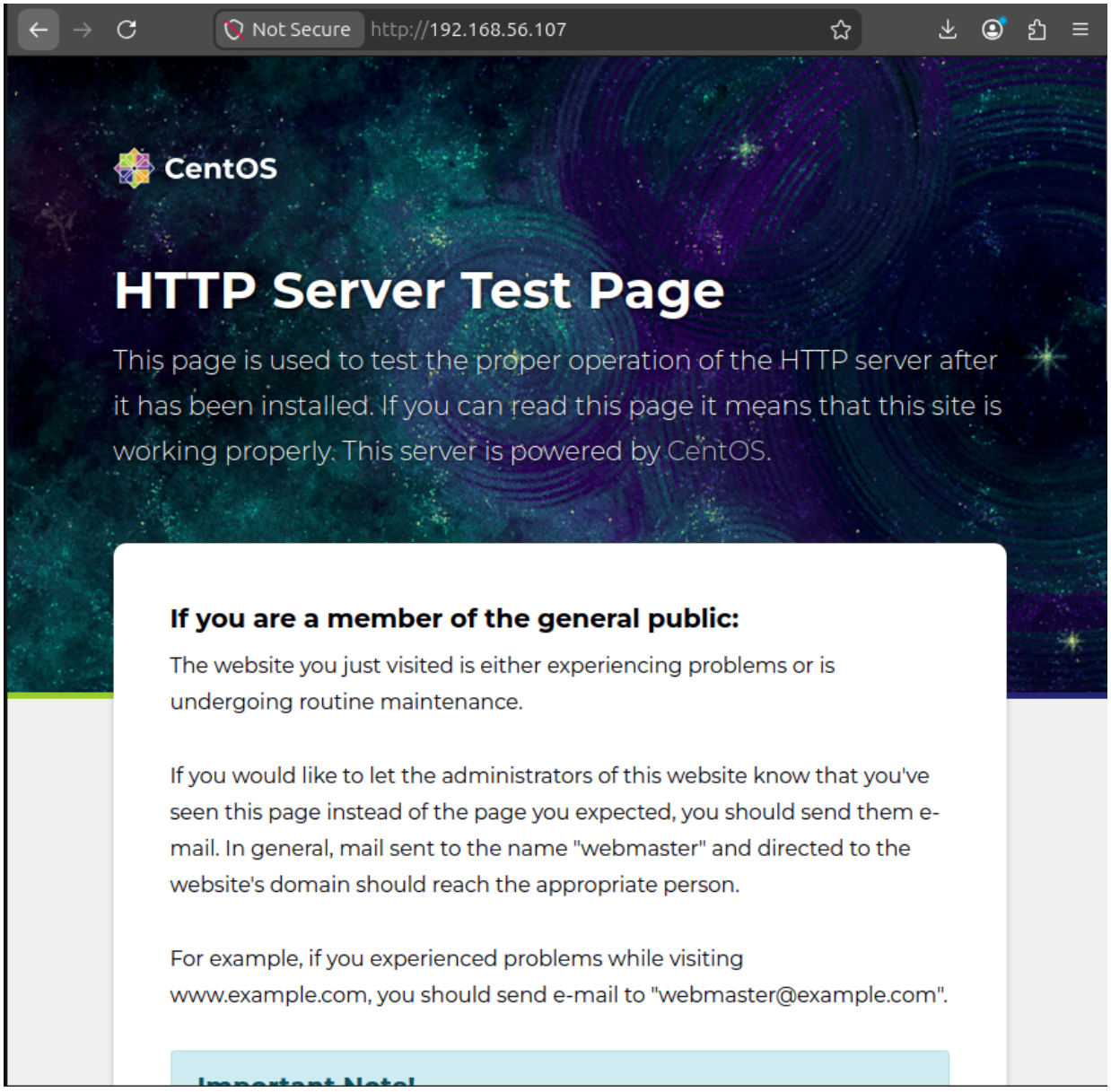
TASK [Gathering Facts] ********************************************************
ok: [192.168.56.106]

PLAY [file_servers] ***********************************************************

TASK [Gathering Facts] ********************************************************
ok: [192.168.56.107]

PLAY RECAP ********************************************************************
192.168.56.104             : ok=4    changed=0    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
192.168.56.106             : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.107             : ok=6    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0

leroy1@leroyque:~/CPE212_QUE$
```



Describe the result.

- In the task mentioned in the playbook, TASK [start httpd (CentOS)] restarts the web service and brings up the CentOS HTTP Server Test Page again. (Note: At this point which is task 3, I have switched into my personal computer in which it changed my ip addresses to be targeted.)

To automatically enable the service every time we run the playbook, use the command enabled: true similar to Figure 7.1.2 and save the playbook.

```
- name: start httpd (CentOS)
  tags: apache, centos, httpd
  service:
    name: httpd
    state: started
    enabled: true
  when: ansible_distribution == "CentOS"
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Supplementary Activity:

1. Creating a Role for Web Servers
2. Creating a Role for Web Servers
3. Creating a Role for Database Servers

```
leroy1@leroyque:~/CPE212_QUE$ ls roles
db  web
```

4. Combining Multiple Roles in a Playbook

Web server playbook:

```
  GNU nano 8.7.1                          roles/web/tasks/main.yml
---
- name: install apache and php for Ubuntu servers
  apt:
    name:
      - apache2
      - libapache2-mod-php
    state: latest
    update_cache: true
  when: ansible_distribution == "Ubuntu"

- name: install apache and php for CentOS servers
  dnf:
    name:
      - httpd
      - php
    state: latest
  when: ansible_distribution == "CentOS"

- name: start httpd (CentOS)
  service:
    name: httpd
    state: started
    enabled: true
  when: ansible_distribution == "CentOS"
```

Database server playbook:

```
  GNU nano 8.7.1                          roles/db/tasks/main.yml
---
- name: install mariadb package (CentOS)
  dnf:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "CentOS"

- name: install mariadb package (Ubuntu)
  apt:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "Ubuntu"

- name: Mariadb - Restarting/Enabling
  service:
    name: mariadb
    state: started
    enabled: true
```

site_roles.yml

```
  GNU nano 8.7.1                                  site_roles.yml
---
- hosts: all
  become: true
  pre_tasks:
    - name: install updates (CentOS)
      dnf:
        update_only: yes
        update_cache: yes
      when: ansible_distribution == "CentOS"

    - name: install updates (Ubuntu)
      apt:
        upgrade: dist
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

- hosts: web_servers
  become: true
  roles:
    - role: web
      tags: web

- hosts: db_servers
  become: true
  roles:
    - role: db
      tags: db
```

To orchestrate multiple roles in a single playbook for different server groups.
Add tags to each role to allow selective execution, e.g., tags: web for webserver,
tags: db for database.

Execution:

```
leroy1@leroyque:~/CPE212_QUE$ ansible-playbook site_roles.yml
BECOME password:

PLAY [all] *********************************************************************************

TASK [Gathering Facts] *********************************************************************
[WARNING]: Host '192.168.56.106' is using the discovered Python interpreter at '/usr/bin/python
3.14', but future installation of another Python interpreter could cause a different interprete
r to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpr
eter_discovery.html for more information.
ok: [192.168.56.106]
[WARNING]: Host '192.168.56.104' is using the discovered Python interpreter at '/usr/bin/python
3.14', but future installation of another Python interpreter could cause a different interprete
r to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpr
eter_discovery.html for more information.
ok: [192.168.56.104]
[WARNING]: Host '192.168.56.107' is using the discovered Python interpreter at '/usr/bin/python
3.9', but future installation of another Python interpreter could cause a different interpreter
 to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpre
ter_discovery.html for more information.
ok: [192.168.56.107]

TASK [install updates (CentOS)] ************************************************************
[WARNING]: Deprecation warnings can be disabled by setting `deprecation_warnings=False` in ansi
ble.cfg.
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to `True` is deprecated, top-level facts wi
ll not be auto injected after the change. This feature will be removed from ansible-core versio
```

```
PLAY RECAP *********************************************************************************
192.168.56.104             : ok=4    changed=0    unreachable=0    failed=0    skipped=3    res
cued=0    ignored=0
192.168.56.106             : ok=4    changed=0    unreachable=0    failed=0    skipped=3    res
cued=0    ignored=0
192.168.56.107             : ok=5    changed=0    unreachable=0    failed=0    skipped=2    res
cued=0    ignored=0
```

–tags web:

```
TASK [Gathering Facts] *********************************************************************
ok: [192.168.56.104]
ok: [192.168.56.106]

TASK [web : install apache and php for Ubuntu servers] *************************************
[WARNING]: Deprecation warnings can be disabled by setting `deprecation_warnings=False` in ansi
ble.cfg.
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to `True` is deprecated, top-level facts wi
ll not be auto injected after the change. This feature will be removed from ansible-core versio
n 2.24.
Origin: /home/leroy1/CPE212_QUE/roles/web/tasks/main.yml:9:9

7     state: latest
8     update_cache: true
9   when: ansible_distribution == "Ubuntu"
          ^ column 9

Use `ansible_facts["fact_name"]` (no `ansible_` prefix) instead.

ok: [192.168.56.106]
ok: [192.168.56.104]

TASK [web : install apache and php for CentOS servers] *************************************
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to `True` is deprecated, top-level facts wi
ll not be auto injected after the change. This feature will be removed from ansible-core versio
n 2.24.
Origin: /home/leroy1/CPE212_QUE/roles/web/tasks/main.yml:17:9

15        - php
16      state: latest
17    when: ansible_distribution == "CentOS"
            ^ column 9

Use `ansible_facts["fact_name"]` (no `ansible_` prefix) instead.

skipping: [192.168.56.104]
```

```
PLAY RECAP *********************************************************************************
192.168.56.104             : ok=3    changed=0    unreachable=0    failed=0    skipped=2    re
cued=0    ignored=0
192.168.56.106             : ok=3    changed=0    unreachable=0    failed=0    skipped=2    re
cued=0    ignored=0
192.168.56.107             : ok=2    changed=0    unreachable=0    failed=0    skipped=0    re
cued=0    ignored=0
```

–tags db:

```
ok: [192.168.56.107]

PLAY [web_servers] *************************************************************************

TASK [Gathering Facts] *********************************************************************
ok: [192.168.56.106]
ok: [192.168.56.104]

PLAY [db_servers] **************************************************************************

TASK [Gathering Facts] *********************************************************************
ok: [192.168.56.107]

TASK [db : install mariadb package (CentOS)] ***********************************************
[WARNING]: Deprecation warnings can be disabled by setting `deprecation_warnings=False` in ansi
ble.cfg.
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to `True` is deprecated, top-level facts wi
ll not be auto injected after the change. This feature will be removed from ansible-core versio
n 2.24.
Origin: /home/leroy1/CPE212_QUE/roles/db/tasks/main.yml:6:9

4     name: mariadb-server
5     state: latest
6   when: ansible_distribution == "CentOS"
          ^ column 9

Use `ansible_facts["fact_name"]` (no `ansible_` prefix) instead.

ok: [192.168.56.107]

TASK [db : install mariadb package (Ubuntu)] ***********************************************
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to `True` is deprecated, top-level facts wi
ll not be auto injected after the change. This feature will be removed from ansible-core versio
n 2.24.
Origin: /home/leroy1/CPE212_QUE/roles/db/tasks/main.yml:12:9

10     name: mariadb-server
```

```
PLAY RECAP *********************************************************************************************
192.168.56.104             : ok=2    changed=0    unreachable=0    failed=0    skipped=0    res
cued=0    ignored=0
192.168.56.106             : ok=2    changed=0    unreachable=0    failed=0    skipped=0    res
cued=0    ignored=0
192.168.56.107             : ok=4    changed=0    unreachable=0    failed=0    skipped=1    res
cued=0    ignored=0
```

All tags:

```
leroy1@leroyque:~/CPE212_QUE$ ansible-playbook site_roles.yml --list-tags

playbook: site_roles.yml

  play #1 (all): all    TAGS: []
      TASK TAGS: []

  play #2 (web_servers): web_servers    TAGS: []
      TASK TAGS: [web]

  play #3 (db_servers): db_servers      TAGS: []
      TASK TAGS: [db]
```

Reflections:

Answer the following:

1. What is the importance of putting our remote servers into groups?
   - The inventory organizes hosts in different groups based on their role, then playbooks refer only to the needed nodes instead of executing all configurations on all machines.
2. What is the importance of tags in playbooks?
   - Application of tags provides selective execution options for the administrators and facilitates the process of running or skipping some tasks to save time while testing or troubleshooting.
3. Why do think some services need to be managed automatically in playbooks?
   - Some Linux distributions such as CentOS do not automatically boot services on installation, so managing them through playbooks provides the possibility of keeping them running and enabled when the server is booting up.

Conclusion:

In this laboratory activity, I gained the knowledge of effectively addressing target nodes by grouping remote servers according to their role in the Ansible inventory file. Furthermore, by separating the playbook into different plays, I was able to create role-specific tasks that enabled the execution of playbooks for specific hosts only, thus omitting unnecessary platforms. Also, I created tags for the task that allowed executing or skipping specific plays without executing the whole playbook. I used Ansible modules for managing service daemons on Linux, therefore, services such as HTTPD have been enabled and started in different distributions automatically. Finally, I organized the tasks in modular roles so as to meet the best requirements for the structure of an Ansible project.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **N/A**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **N/A**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **N/A**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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